



Evaluation of a model for sound radiation from the vocal tract wall

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Abstract

This study evaluated a simulation model for sound radiation from the vocal tract wall in the context of articulatory speech synthesis. In this model, the vocal tract is represented in terms of incremental contiguous tube sections, where the wall of each section reacts to the local sound pressure like a damped spring-mass systems. The surface-radiated sound from this model was compared with that of six real speakers by means of the voicebars of /b,d,g/ in different vowel contexts. Based on this, the wall parameters of the real speakers were estimated by fitting the simulated to the real voicebars. The parameter optimization allowed a close reproduction of the natural voicebar spectra with root-mean-square errors between 2.26 dB and 3.82 dB in the frequency range from 0–800 Hz. Hence, despite its simplicity, the modeling method is very well suited for articulatory speech synthesis.

Index Terms: voicebar, sound radiation from skin, articulatory synthesis

1. Introduction

Articulatory speech synthesizers generate speech based on articulatory models of the speech production system and physics-based acoustical and aerodynamic simulations. In order for the speech to sound natural, the simulation models should take into account all relevant features of the real speech production system. One important feature of the vocal tract is its soft wall, which vibrates in response to the sound pressure within the vocal tract. This has multiple effects on the produced speech:

- The interaction of the sound field in the vocal tract with the soft walls increases the frequencies and bandwidths of the lower vocal tract resonances, and hence causes an acoustic energy loss [1, 2, 3].
- The elasticity of the walls supports the maintenance of phonation during voiced plosives (passive cavity enlargement [4]).
- The vibrating wall radiates sound, which adds to the sound radiated from the mouth and nostrils. This sound is visible as a “voicebar” in the spectrogram of voiced plosives, but is also a “hidden” component of other speech sounds.

These effects have been taken into account to different degrees in articulatory synthesizers. With regard to the effect on the resonances, most acoustical simulations model soft walls as a *distributed* wall impedance [5, 6, 7, 8, 9, 10, 11, 12, 13, 14], while others model it as a *lumped* impedance located at the larynx [15, 16] or omit it entirely [17]. Of the synthesizers that include wall losses, only a few also simulate the sound that is radiated from the skin [14, 5, 6, 7, 9, 10]. Here, [14, 5] consider the radiation by all vocal tract wall sections, while the others

consider sound radiation only in the vicinity of the larynx. The most complete simulations are those that consider a distributed wall impedance and a distributed radiation from the vibrating wall [14, 5].

All mentioned studies model the wall impedance as that of a damped spring-mass system. The mechanical impedance Z_w per unit area of this system is

$$Z_w(\omega) = P(\omega)/V(\omega) = K_w/(j\omega) + j\omega M_w + B_w,$$

where ω is the circular frequency, $j = \sqrt{-1}$, $P(\omega)$ is the pressure acting on the walls, $V(\omega)$ is the wall velocity in response to this pressure, K_w is the stiffness per unit area, M_w is the wall mass per unit area, and B_w is the damping per unit area. The values of K_w , M_w , and B_w are in reality different for different surfaces within the vocal tract, e.g. the hard palate versus the tongue. In all simulation models, however, they are treated as (potentially speaker-specific) constants, whose values vary from model to model. Possible ranges of variation of these parameters are given by Stevens [3] as

$$K_w: 30\,000 \dots 100\,000 \text{ dyn/cm}^3,$$

$$M_w: 1.0 \dots 2.0 \text{ g/cm}^2,$$

$$B_w: 800 \dots 2000 \text{ dyn-s/cm}^3.$$

To simulate the sound radiated from the vocal tract wall, the local wall velocity can be integrated over the wall surface area to obtain a corresponding volume velocity source. However, so far it is unclear how realistic the sound radiated from this fairly simple model is. Therefore, the goal of this study was to explore

- whether the wall model is sufficiently detailed to reproduce the characteristic acoustic features of the sound radiated from the skin of human subjects, and
- to what extent it is possible to estimate the wall parameters K_w , M_w , and B_w from the sound radiated from the skin.

The sound emitted from the vocal tract wall was analyzed by means of the voicebars of voiced plosives.

2. Method

Section 2.1 discusses the analysis of voicebars of human subjects, and Section 2.2 describes the synthesis of voicebars in the articulatory speech synthesizer VocalTractLab [18]. Section 2.3 presents the method to estimate the wall parameters.

2.1. Analysis of real voicebars

High-quality speech signals with a sampling rate of 44 100 Hz were recorded from six male native German speakers (23–46 years) in a soundproof room using a studio microphone (M930 by Microtech Gefell), a USB audio interface (UH-7000 by TASCAM), and a laptop running the software Audac-

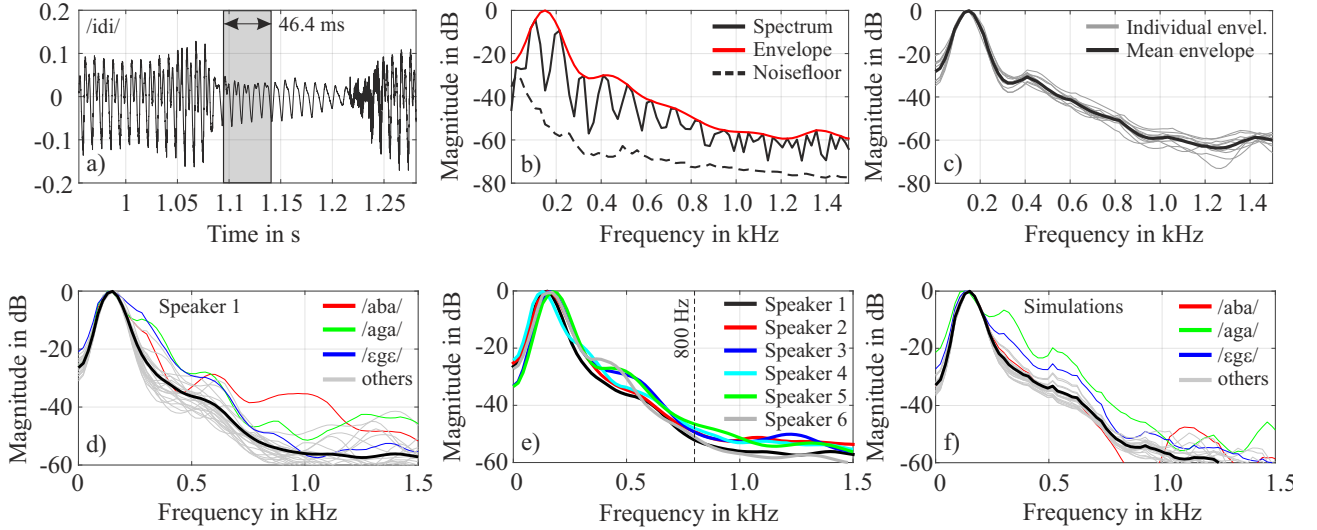


Figure 1: Analysis of real voicebars. a) Audio waveform of /idi/ of speaker 1 with the selected initial part of the voicebar for /d/ (gray). b) Spectrum and envelope of the voicebar in a), and the noise floor. c) All 10 voicebar envelopes of /idi/ of speaker 1 (gray) and their mean (black). d) Mean voicebar envelopes of all 24 VCV items of speaker 1 (thin lines) and their mean (thick black line). e) Mean voicebar envelopes of all 6 speakers. f) Voicebar envelopes of all 24 simulated VCV items (thin lines) and their mean (thick black line). Here, the wall and voice source model parameters of speaker 1 were used.

ity 2.3.3 (<http://audacity.sourceforge.net/>). Female speakers were not recorded in this experiment, because the vocal tract model for the simulations was only available for a male German speaker. Each speaker produced repeated consonant-vowel (CV) sequences like /bababa.../ for all combinations of the 8 tense German vowels /a, e, i, o, u, ε, ø, y/ and the 3 voiced plosives /b, d, g/ for a total of 24 /CVCVC.../ sequences. All sequences were spoken slowly, with flat intonation and with 10–20 plosives per sequence (typically with one or two breathing pauses). The speakers were standing in front of the microphone at a distance of about 40 cm.

In each sequence, the initial parts of the voicebars of 10 of the plosives were extracted, as shown for /idi/ in Figure 1a. The extracted voicebar sections had a length of 46.4 ms (=2048 audio samples). Only plosives with a sufficiently long voicebar were used.

For each extracted voicebar, the “true spectral envelope”, which is based on cepstral smoothing, was calculated according to [19] and peak-normalized to 0 dB. The lifter $w(\tau)$ (window in the quefrequency domain) was designed as

$$w(\tau) = \begin{cases} 1, & \tau \leq \tau_1 \\ \frac{1}{2} + \frac{1}{2} \cos(\pi(\tau - \tau_1)/(\tau_2 - \tau_1)), & \tau_1 < \tau \leq \tau_2 \\ 0, & \text{otherwise,} \end{cases}$$

where τ is the quefrequency, $\tau_1 = 0.4 \cdot T_0$, $\tau_2 = 0.8 \cdot T_0$, and T_0 is the fundamental period of the voice signal, which was estimated by means of the autocorrelation function. This lifter generates a spectral envelope which passes nicely through the harmonic peaks of the magnitude spectrum of the voicebar segment, as shown in Figure 1b. The figure also shows that the energy of the voicebars is mainly concentrated at low frequencies and that there is a strong spectral decay towards higher frequencies. A screening of the audio recordings showed that the spectral voicebar magnitude was at least 20 dB above the noise floor up to $f_{\max} = 800$ Hz in the majority of cases. Since the spectra were increasingly affected by noise above this frequency, $f_{\max}$

was taken as the upper frequency limit for the subsequent analysis.

For each /CVCVC.../ sequence, the spectral envelopes (SEs) of the voicebars of all 10 consonants were extracted and averaged, as shown for /dididi.../ of speaker 1 in Figure 1c. This resulted in 24 envelopes (one for each CV combination) for each speaker, which are shown for speaker 1 as the thin lines in Figure 1d. The envelopes are relatively similar for most CV combinations, but interestingly, a few envelopes stand out for all speakers, namely for the CV combinations /aba/, /aga/, and /εge/ (red, green, and blue curves).

In order to obtain one final representative SE per speaker, the 24 envelopes for the different CV combinations were averaged. They are shown in Figure 1e for all six speakers. We note that the envelopes are relatively similar for all speakers (more similar than for the different CV combinations of the same speaker) and have a characteristic “knee” around 500 Hz.

2.2. Physical simulation of voicebars

Simulations were performed with an extended version of the articulatory speech synthesizer VocalTractLab 2.3 (www.vocaltractlab.de, [20, 18, 21]). The components of the synthesizer are illustrated in Figure 2. A geometric model of the vocal folds [22] and a geometric model of a male adult vocal tract [18] are transformed into a tube model that represents the airways by a series of short cylindrical tube sections, including branches for the nasal cavity and the piriform fossae (not shown) [7]. This tube model is equivalent to an acoustic network, where each tube section corresponds to a two-port network with lumped elements (Figure 2e). The network is driven by a pressure source P_{sub} at the tracheal end and is terminated with a radiation impedance Z_{rad} at the mouth and nostrils. The two-port network for each tube section i contains acoustic resistors R_i , acoustic inductances L_i , and the acoustic compliance C_i [23]. According to [5, 8, 24], the effect of the soft wall is modeled by a branch with the network elements $R_{w,i}$, $L_{w,i}$,

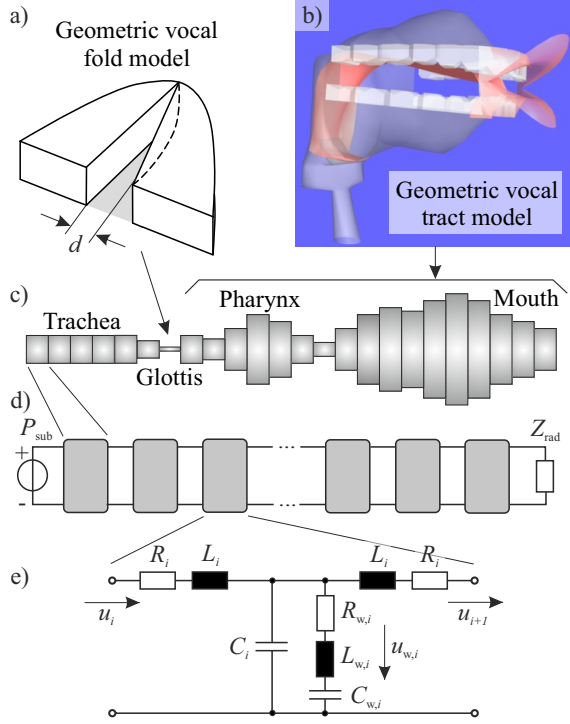


Figure 2: Overview of the simulation model. a) Vocal fold model. b) Vocal tract model. c) Tube model. d) Transmission-line model. e) Two-port network for one tube section.

$C_{w,i}$, which are directly related to the mechanical parameters of the vocal tract wall introduced before:

$$\begin{aligned} R_{w,i} &= B_w / (S_i \cdot l_i) \\ L_{w,i} &= M_w / (S_i \cdot l_i) \\ C_{w,i} &= (S_i \cdot l_i) / K_w. \end{aligned}$$

Here, l_i and S_i are the length and circumference of tube section i , respectively. The volume velocity $u_{w,i}$ through this branch of the circuit equals the wall velocity (perpendicular to the surface) times the surface area of the tube section. The sound pressure radiated from the vibrating wall of the whole vocal tract was modelled as the sum $\sum_i u_{w,i}$ across all tube sections filtered by the radiation characteristic. The radiation characteristic was assumed to be the same as for sound radiated from the mouth and nostrils using the realistic FEM-based filter obtained in [25].

To generate connected utterances, the synthesizer implements the concept of a gestural score, which specifies an utterance in the form of the articulatory gestures that constitute it [26]. These scores were used to synthesize /VCV/ utterances for all combinations of the vowels /a, e, i, o, u, ε, ø, y/ and the consonants /b, d, g/, i.e. for the same CV combinations as in Section 2.1. In these synthetic utterances, the voicebars were extracted and processed in the same way as for the natural recordings to obtain one spectral envelope for each of the 24 CV combinations. Figure 1f shows these envelopes for synthesis settings optimized for subject 1 (see below). Both the individual SEs (thin lines) and the average SE (thick black line) are very similar to those obtained from the natural utterances of subject 1 in Figure 1d. Not only the “knee” in the curve at around 600 Hz and -40 dB appears in the simulated data, but also the “outlier” voicebars of /aba/, /aga/, and /εgε/.

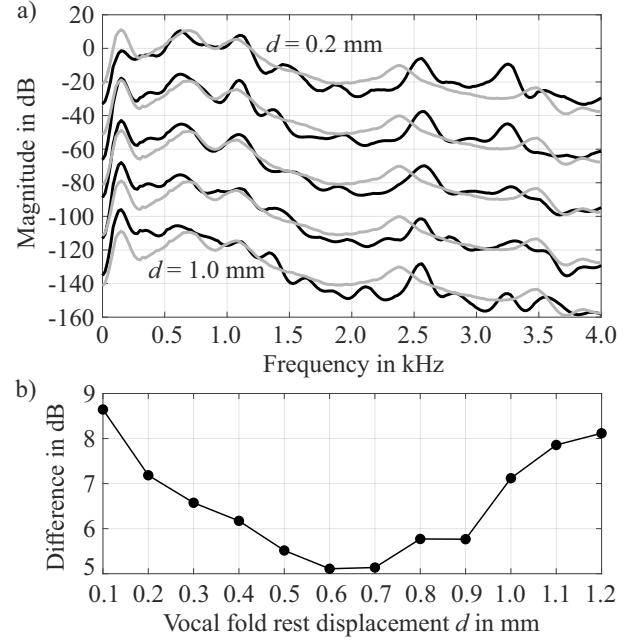


Figure 3: a) Spectral envelopes of synthetic /a/ for different vocal fold rest displacements (black) compared to the average /a/ envelope of speaker 1 (gray). b) Root-mean-square difference between the synthetic and natural /a/ envelopes in dB for different vocal fold rest displacements.

2.3. Determination of vocal tract wall parameters

To estimate the wall parameters of the six subjects, we used the method of “analysis by synthesis”. For this, it is important to understand that a voicebar spectrum is affected both by the sound pressure that drives the wall vibration inside the vocal tract, and by the mechanical wall parameters. The spectral composition of the sound pressure in the vocal tract is in turn determined by the glottal source signal and the shape of the vocal tract. Hence, to estimate the wall parameters, the simulated source signal and vocal tract shape must correspond to that of the subject. With regard to the vocal tract shape, the simulated voicebars were extracted from the same CV combinations as the real voicebars. Although the vocal tract model does not perfectly correspond to that of any of the six subjects, it still assumed the shape for the same plosives in the same vowel contexts, which should be sufficiently similar with regard to the considered frequency range up to 800 Hz.

With regard to the glottal source signal, the f_0 in the simulations was set to the average f_0 during the voicebars of the corresponding subject. The other major feature of the glottal source signal is its spectral tilt, which is related to the degree of abduction of the vocal folds. In the vocal fold model [22] it is controlled by the glottal rest displacement of the upper and lower vocal fold edges from the glottal midline at the posterior end (d in Figure 2a). This parameter was individually estimated for each subject by fitting the spectral envelope of the synthesized vowel /a/ to that in the natural speech recordings. The estimation of d using /a/ instead of the voicebars themselves is more robust due to the broader frequency spectrum of /a/. The spectral envelopes of the natural and synthetic vowels were determined as in Section 2.1. For each subject, the envelopes of 10 naturally produced /a/’s were averaged to obtain a representa-

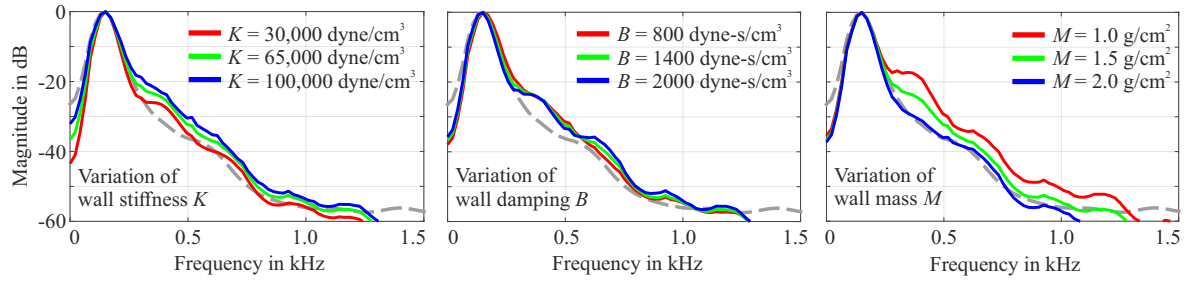


Figure 4: Variation of the simulated voicebar envelopes of speaker 1 for different vocal tract wall parameters values. The parameters that were not varied were set to their optimal values (see Table 1). The gray dashed line is the real voicebar SE of speaker 1.

tive envelope. Figure 3a illustrates how the synthetic SEs (black curves) compares to the real envelope of subject 1 (gray curves) for increasing simulated glottal rest displacements d . Here, for $d = 0.2$ mm the synthetic envelope has too small amplitudes at low frequencies and too high amplitudes at high frequencies, i.e. the synthetic voice is too pressed. For $d = 1.0$ mm it is the other way around, i.e. the synthetic voice is too breathy. As shown in Figure 3b, a minimal root-mean-square (RMS) difference between the envelopes for the range from 0 to 4 kHz was found for $d = 0.6$ mm, which was taken as the best estimate for subject 1. The d value for the other speakers was estimated in the same way.

With the best estimates for the source parameters f_0 and d for each subject, the wall parameters were estimated by grid search for all combinations of $K_w \in \{30\,000, 65\,000, 100\,000\}$ dyn/cm³, $M_w \in \{1.0, 1.5, 2.0\}$ g/cm², and $B_w \in \{800, 1400, 2000\}$ dyn-s/cm³. The values are the minima and maxima for these parameters provided by Stevens [3] with one additional value in the middle of the range. A finer gradation of values was not considered necessary, because the effect of the changes on the voicebars is relatively small, as shown in Figure 4. For each wall parameter combination, the average SE of the synthesized voicebars for all 24 CV combinations was compared with the average SE of the subject by means of the RMS difference between 0 and 800 Hz.

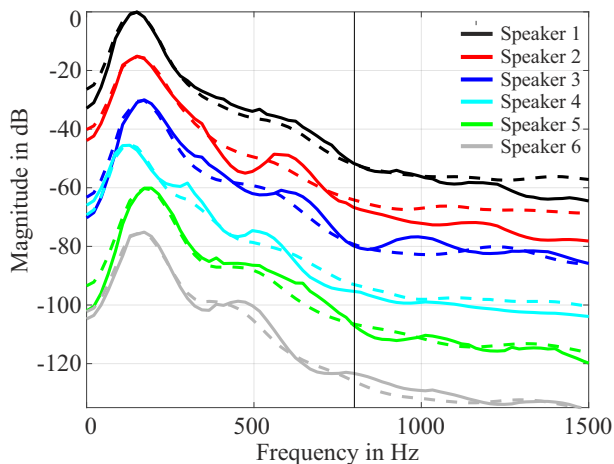


Figure 5: Voicebar spectral envelopes of the subjects (dashed lines) versus the matched synthetic spectral envelopes (solid lines). The vertical black line at 800 Hz marks the frequency up to which the spectra were matched.

Table 1: Estimated parameters for the subjects 1–6.

#	f_0 (Hz)	d (mm)	K_w (dyn/cm ³)	B_w (dyn-s/cm ³)	M_w (g/cm ²)	RMSE (dB)
1	103	0.6	100 000	1400	2.0	2.62
2	111	0.8	65 000	800	2.0	2.99
3	115	0.2	100 000	2000	2.0	3.46
4	101	0.9	65 000	1400	2.0	2.92
5	127	0.3	100 000	800	2.0	3.82
6	130	0.6	100 000	2000	2.0	2.26

3. Results and conclusions

The estimated voice source parameters f_0 and d and the wall parameters K_w , B_w , and M_w are given in Table 1 for all subjects, together with the RMS error between the envelopes of the synthetic and natural voicebars. The results for d indicate that subjects 2 and 4 had a rather breathy voice (big vocal fold rest displacement), subjects 3 and 5 had a more pressed voice, and the voices of subjects 1 and 6 were in between. With regard to the wall parameters, the values for K_w are either the middle or maximum value of the range proposed by Stevens [3], while the values for B_w vary across the whole range. The wall mass M_w was 2.0 g/cm² for all subjects, which is the upper limit of the proposed M_w range. The re-synthesis of the natural voicebars worked well with RMS errors ranging from 2.26 to 3.82 dB. The match is shown in Figure 5, where the SEs of the synthetic voicebars (solid lines) are overlaid with the corresponding SEs of the subjects (dashed lines).

From these results we can conclude that the presented model for sound radiation from the vocal tract wall is quite realistic and allows an accurate reproduction of the voicebars in real speech. While this model has been proposed a long time ago [5], it was not previously evaluated by comparison with real speech. For the mechanical wall parameters, it seems to be an acceptable simplification to use uniform values for the entire vocal tract rather than different values for different parts of the wall (e.g. the hard versus the soft palate). For M_w , 2.0 g/cm² seems to be a representative value, which corresponds to an average wall thickness of 2 cm. The value for B_w seems to be more speaker-specific. While M_w and B_w can be assumed constant for a particular speaker, the stiffness K_w controls the compliance of the wall and can be used to adjust the voicebar duration during voiced plosives in articulatory speech synthesis.

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5. References

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